THE WHITE ARMY

A CASE ON PARAPARESIS

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DEMOGRAPHIC DETAILS AND CHIEF COMPLAINTS

A **78** year old **gentleman** from a **rural area** in Shikaripura (about 20 Kilometers away from health care setup), a former **farmer** who now does his routine physical activity at home, came to our outpatient Department on Wheelchair with chief complaints of

1. **Weakness of bilateral lower limbs since 15 days**
2. **Numbness of distal lower limbs since 15 days**

He is a **right handed** individual.

The history was obtained from the patient and his son.

I. HISTORY OF PRESENTING ILLNESS

The patient was apparently well in his usual state of health. He could walk around without assistance and do his daily activities by himself. About 15 days ago, he found difficulty in walking which he told was because his ‘legs were weak’. He could walk a certain distance after which his legs would buckle and he would need assistance of one person to walk. He says that this happened every time he tried to walk, with ‘the walking distance without assistance’ being approximately the same every time. He also reported that his chappals would slip from his legs and he wouldn’t be aware of it. He reported that the weakness in both his lower limbs started at the same time.



Since about 5-6 days, he has been finding it difficult to get up from the squatting position, needing assistance, but he couldn’t tell if he could feel the presence of his clothes above the level of his ankles bilaterally. Thus, he has a 15 day history of bilateral lower limb weakness which started off insidiously as a distal weakness but now has proximal muscle weakness too (Distal>Proximal)

I

History of Bilateral sensory deficits (feeling numb), since about 15 days, onset being at the same time/slightly later than the motor weakness with ? Proximal progression with no history of tingling/burning/pins and needles sensation. He couldn’t tell if there was loss of pain and temperature in the area of numbness.



There is no history of difficulty gripping objects with the upper limb or doing overhead activities like combing. No sensory deficits in the upper limbs

No history of breathlessness, orthopnea

No History of difficulty holding his neck in upright position

No history suggestive of cranial nerve deficits

No history of swaying while walking

No history of syncope on standing, palpitations



No history of stiffness of the limbs/flexor spasms

No history of bowel or bladder incontinence

No history of involuntary movements in the thigh/gluteal muscles

No history of root pain



No history of back ache at the onset of symptoms/through the course of the illness

No history of trauma to the back/fall from height

No history of fever

No history of preceding cough/coryzal symptoms, diarrhea

No history of recent vaccinations

II. HISTORY OF PAST ILLNESS

- History of an episode exactly like the current history, about 6 months back, preceded by fever and coryzal complaints. Symptoms lasted for about 2 weeks and resolved completely following treatment in a tertiary care setup. Symptoms were completely absent post this episode. He was given a tablet for nerve strength for 2 months.

II

- History of **diabetes mellitus** since 15 years. He was on TAB METFORMIN for about 14 years but was added on INSULIN once daily at night with TAB METFORMIN after his sugars were not controlled. About 2.5 months ago, he was found unconscious in his room and was taken to the hospital, was told to have had a hypoglycemic episode. Post this, INSULIN was stopped. Sugars were last checked during routine checkup about a month back and were recorded to be 400+. He has been on INSULIN+METFORMIN ever since.

II

- History of **hypertension** since 1 year on regular once daily tablet. His BP was last checked about a month ago and was within the normal range
- No history of Tuberculosis, Bronchial asthma, Ischemic heart disease, Childhood epilepsy
- History of cataract surgery in both eyes 6 years ago.

III. TREATMENT HISTORY

- On **TAB METFORMIN** 500 mg BD plus **INSULIN** injections.
- On **TAB RAMIPRIL** 5 mg OD

IV. PERSONAL AND SOCIAL HISTORY

- He follows a vegetarian diet and has good appetite
- His sleep is undisturbed and has normal pattern.
- He has normal bowel and bladder functions
- There is no history of intake of alcohol or tobacco in any form.
- No history of loss of weight
- He holds the Hospital Arogya Card but redeems no health benefits from Government schemes.

V. FAMILY HISTORY

- He lives with his son, who is the primary care taker.
- There is no family history of similar complaints

SUMMARY OF MY CASE

An elderly male, known diabetic (uncontrolled) and hypertensive, comes with a 15 day history of weakness in both his lower limbs, with distal to proximal progression (Distal>Proximal) and feeling numb in his bilateral lower limbs. No history suggestive of involvement of upper limb, cranial nerves, cerebellum, spinal cord or anterior horn cells. History of similar episode 6 months back. I am thinking of an ACUTE POLYNEUROPATHY

EXAMINATION FINDINGS

I.GENERAL PHYSICAL EXAMINATION

The patient was examined on Day 1 of admission.

He was lying down comfortably on his bed.

He was conscious and co-operative and was well oriented to time, place and person.

He was moderately built and poorly nourished with a body mass index of 17.9

Vitals

- **Pulse:** 104 beats per minute, regular rhythm, high volume, no special character; Arterial wall is not thickened; Peripheral pulses are felt with normal intensity bilaterally; There is no radio-radial delay or radio-femoral delay.
- **Blood pressure:** 140/64 mm of Hg, right arm, supine position.
- **Respiratory rate:** 18 cycles per minute, predominantly abdominothoracic, pattern and depth is normal and there is no use of accessory muscles for respiration.
- Patient was **afebrile** (Temporal temperature of 37.3 degree)

Head to toe examination:

Pallor noted in lower palpebral conjunctiva, oral mucosa and nail bed; Palmar creases looked normal.

No icterus, clubbing, cyanosis, generalised lymphadenopathy, pedal edema.

No signs of nutrition deficiencies

No neurocutaneous markers; No hypo/hyperpigmented macules

No thickening of peripheral nerves

II. NERVOUS SYSTEM EXAMINATION

A. Higher Mental functions: MMSE = 30

- Right handed individual. Has received primary education
- Conscious, coherent, co-operative
- Well oriented to time, place and person
- Appearance and Behaviour: Appropriate
- Emotionally stable
- General intelligence: Average
- Recent, immediate, remote memory intact
- Speech: Comprehension, Spontaneous speech, naming, reading, writing - Normal

B. Cranial nerve examination

CN	RIGHT	LEFT
I	Normal	Normal
II	**Visual acuity:** Snellen chart not available. Counting finger at 6 metre **Visual field:** Normal by confrontation technique **Colour vision:** Normal	**Visual acuity:** Snellen chart not available. Counting finger at 6 metre **Visual field:** Normal by confrontation technique **Colour vision:** Normal
III, IV, VI	**Eyelids:** No ptosis **Eyeballs at rest:** No exophthalmos/ enophthalmos, no strabismus **Extra-ocular muscles:** Motility normal **Pupils:** Round, reactive, no RAPD **Nystagmus:** Absent	**Eyelids:** No ptosis **Eyeballs at rest:** No exophthalmos/enophthalmos, no strabismus **Extra-ocular muscles:** Motility normal **Pupils:** Round, reactive, no RAPD **Nystagmus:** Absent

II. B

V	**Sensory:** Intact **Motor**: Intact **Reflexes:** Corneal and conjunctival intact	**Sensory:** Intact **Motor:** Intact **Reflexes:** Corneal and conjunctival intact
VII	**Motor:** Normal **Taste sensation (**in anterior one third of the tongue): Normal **Reflexes:** Corneal and conjunctival intact	**Motor:** Normal **Taste sensation (**in anterior one third of the tongue): Normal **Reflexes:** Corneal and conjunctival intact
VIII	**Cochlear:** Rinne negative, Weber not lateralized, ABC positive **Vestibular:** Deferred	**Cochlear:** Rinne negative, Weber not lateralized, ABC positive **Vestibular:** Deferred

II. B

IX and X	**Motor**: Normal **Sensory:** Normal **Reflex:** Gag reflex Intact	**Motor**: Normal **Sensory:** Normal **Reflex:** Gag reflex Intact
XI	Normal power of sternocleidomastoid and trapezius	Normal power of sternocleidomastoid and trapezius
XII	No deviation on protrusion; No atrophy, fasciculation	No deviation on protrusion; No atrophy, fasciculation

Impression: All cranial nerves intact bilaterally

C. Motor system examination

A. Attitude of limbs

Upper limbs: Shoulder adducted and in neutral flexion and rotation, elbow extended, wrist flexed, finger flexed

Lower limbs: Hips adducted and in neutral flexion and rotation, knees extended, ankle plantarflexed

B. Bulk of the muscles

Parameter	LEFT	RIGHT
Arm circumference	16cm	16cm
Forearm	12cm	12cm
Thigh	25cm	25cm
Calf	20cm	20cm

Impression: No evidence of atrophy of group of muscles,

No atrophy of small muscles of the hand

III C

C. Tone of the muscles

	LEFT	RIGHT
A. Upper limb i. Flexor	Normal	Normal
ii. Extensor	Normal	Normal
B. Lower limb i. Flexor	Normal	Normal
ii. Extensor	Normal	Normal

Impression: No evidence of hypotonia or hypertonia

III C

D. Power

Parameter	LEFT	RIGHT
1. Movement at shoulder joint		
Flexion	5	5
Extension	5	5
Abduction	5	5
Adduction	5	5
2. Movement at elbow joint		
Flexion	4+	4+
Extension	4+	4+
3. Movement at wrist joint		
Flexion	4+	4+
Extension	4+	4+
Abduction	4+	4+
Adduction	4+	4+
4. Hand grip	Strong	Strong

III C

D. Power

5. Movement at Hip joint		
A. Flexion	4-	4-
B. Extension	4-	4-
C. Abduction	4-	4-
D. Adduction	4-	4-
6. Movement at knee joint		
A. Flexion	4-	4-
B. Extension	4-	4-

7. Movement at ankle joint		
A. Plantar flexion	3	3
B. Dorsiflexion	3	3
8. Toe movements		
A. Flexion	3	3
B. Extension	3	3

Impression: Power is reduced in both lower limbs (Distal>Proximal) and distal upper limbs (Lower limbs>Upper limbs)

III C

E. Co-ordination:

A. Finger Nose Test: Able to perform

B. Knee- heel test: Able to perform

F. Involuntary movements: Bilateral fine tremors in the upper limbs, No fasciculations seen

III C

G. Reflexes

REFLEX	LEFT	RIGHT
Superficial Reflex		
A. Corneal reflex	Present	Present
B. Abdominal reflex	Present	Present
C. Cremasteric reflex	Not done	Not done
D. Plantar Reflex	**No response**	**No response**

Deep reflex		
A. Biceps reflex	2+	2+
B. Supinator jerk	1+	1+
C. Triceps jerk	1+	1+
D. Knee reflex	**0***	**0***
E. Ankle reflex	**0***	**0***

*: With reinforcement

Impression: Bilaterally plantar reflex is absent; Deep tendon reflexes absent in bilateral lower limbs

H. Pronator drift:

III

D. Sensory system examination:

PARAMETERS	RIGHT	LEFT
Spinothalamic sensations A. Pain B. Temperature C. Deep touch D. Pressure	Absent below the elbow (Anterior and posterior) and in entire lower limbs (anterior and posterior). Perineum was not checked.	Absent below the elbow (Anterior and posterior) and in entire lower limbs (anterior and posterior). Perineum was not checked.

III D

Posterior column sensations		
A. Fine touch	Absent below the elbow (Anterior and posterior) and in entire lower limbs (anterior and posterior). Perineum was not checked.	Absent below the elbow (Anterior and posterior) and in entire lower limbs (anterior and posterior). Perineum was not checked.
B. Vibration	**Absent in lower limbs**	**Absent in lower limbs**
C. Joint sense	Intact	Intact
D. Position sense	Intact	Intact

III D

Cortical sensations		
A. Tactile Localization	Intact	Intact
B. Two point discrimination	Intact	Intact
C. Stereognosis	Cannot be commented on	Cannot be commented on
D. Graphesthesia	Intact	Intact
E. Sensory inattention	-	-

Rombergs sign: Couldn’t be elicited because he told he cannot stand

Impression: All sensations Impaired in upper limbs below elbow and lower limbs below hip

Except joint and position sense in all limbs and vibration sense in upper limbs

III

E. Stance and gait: Couldn’t be commented on

F. No signs of cerebellar dysfunction: Titubation, scanning speech, impaired coordination, hypotonia, Dysdiadochokinesia, nystagmus, penduclar knee jerk

G. No signs of autonomic dysfunction

H. No signs of meningeal irritation

REVIEW OF OTHER SYSTEMS

Respiratory system examination:

Bilateral normal vesicular breath sounds heard over all lung areas

Gastrointestinal system examination:

Abdomen soft, non-tender; No evidence of organomegaly

Cardiovascular system examination:

S1, S2 heard. No abnormal splitting. No added sounds/murmurs

DIAGNOSIS

- **Clinical:** Paraparesis with distal sensory deficit
- **Anatomical:** Peripheral nerves, axonal
- **Etiological:** Diabetes

INVESTIGATIONS

1. Complete blood picture: Hb
2. Blood sugars
3. Vitamin B12 levels
4. Serum urea and creatinine
5. Nerve conduction studies


THANK YOU